

Retail Sales & Customer Intelligence Analytics using Data Analytics and AI

IBM SkillsBuild Academic Internship – Data Analytics with AI

Capstone Project Report

Submitted by: Aditya Sharma

Program: Data Analytics with AI | Institution: Graphic Era Hill University

Abstract

This capstone project presents a complete end-to-end data analytics and artificial intelligence solution applied to the Tableau Sample Superstore retail transaction dataset. The project spans the full analytics lifecycle: data ingestion and quality assessment, cleaning and feature engineering, exploratory data analysis (EDA), SQL-based business intelligence queries, customer segmentation using RFM (Recency-Frequency-Monetary) methodology, product profitability analysis, time-series sales forecasting using machine learning, and AI-assisted natural-language business insights. All analytical results are derived entirely from the actual dataset without fabrication. Key findings include total revenue of \$2,326,534 across 5,111 orders, a portfolio profit margin of 12.56%, identification of loss-making product sub-categories (Tables, Bookcases, Supplies), and a strong negative correlation between discount depth and profitability. The project is delivered as a modular Python package, an interactive multi-page Streamlit dashboard, and a structured SQL analysis layer. This report documents the methodology, findings, limitations, and responsible AI considerations for each component.

1. Introduction

The retail industry generates vast volumes of transactional data that, when properly analysed, can yield actionable intelligence across sales strategy, customer relationship management, inventory optimisation, and financial planning. Data analytics and artificial intelligence are increasingly central to competitive advantage in this sector, enabling organisations to move beyond descriptive reporting toward predictive and prescriptive decision-making.

This project was developed as part of the IBM SkillsBuild Academic Internship in Data Analytics with AI. The objective is to demonstrate proficiency across the complete analytics pipeline using a realistic, widely referenced retail dataset. The project integrates traditional statistical analysis, SQL-based business intelligence, machine learning for forecasting, and a generative AI module for natural-language interpretation of computed findings.

All code is written in Python 3.10+ using a modular, reproducible architecture. An interactive Streamlit dashboard provides a professional interface for business stakeholders to explore the analytical outputs without requiring technical expertise. The entire project is designed to be reproducible from a clean environment using only the provided instructions.

2. Business Problem

A multi-category retail operation faces several interconnected business challenges that analytics can address:

- **Revenue and profitability visibility:** Management requires a clear understanding of which product categories, sub-categories, geographic regions, and customer segments are generating or eroding profit. Simple revenue reporting is insufficient because high-revenue products may still be unprofitable due to discounting or operational costs.
- **Discount strategy:** The business applies variable discounts across products and segments. Without analysis, it is unclear whether heavy discounting is driving volume growth that compensates for margin compression, or whether it is simply destroying value without proportional volume benefit.
- **Customer intelligence:** The customer base is large and heterogeneous. Without segmentation, marketing and sales resources cannot be efficiently allocated to high-value or at-risk customers.
- **Sales forecasting:** Operational planning requires forward-looking sales estimates. A data-driven forecasting model, even a simple one, provides more defensible projections than intuitive guesses.
- **Insight accessibility:** Analytical findings must be communicable to business stakeholders who may not possess technical data literacy. Natural-language AI interpretation bridges this gap.

3. Objectives

The project objectives are:

- **O1:** Ingest, assess, and clean the Superstore dataset with full documentation of data quality issues and cleaning decisions.
- **O2:** Engineer analytically useful features (temporal decomposition, profit margin, sales per unit) without modifying the raw data.
- **O3:** Conduct comprehensive EDA covering sales trends, profitability, discounting behaviour, and geographic performance.
- **O4:** Build a structured SQL analysis layer in SQLite with business-meaningful queries addressing the key business questions.
- **O5:** Implement RFM-based customer segmentation to identify customer value tiers.
- **O6:** Analyse product-level profitability to identify high-value and loss-making products.
- **O7:** Build and evaluate time-series sales forecasting models with proper temporal train/test splits.

- O8: Integrate a responsible generative AI module that interprets verified analytical results in natural language.
- O9: Deliver an interactive Streamlit dashboard accessible to non-technical stakeholders.

4. Dataset Description

4.1 Source

The Tableau Sample Superstore dataset is a widely used demonstration dataset originally published by Tableau Software. It represents a fictional but realistic multi-category US retail operation selling Furniture, Office Supplies, and Technology products to Consumer, Corporate, and Home Office customer segments. The dataset is commonly used in academic and professional analytics training due to its breadth, referential completeness, and the diversity of analytical questions it supports. The file used in this project is sample_
_superstore.xls.

4.2 Dataset Structure

After loading and inspection, the dataset presents the following characteristics:

Attribute	Value	Notes
Total rows	10,194	Individual order line items
Total columns	21	After initial load
Date range	2023 – 2026	4 full years of transactions
Unique orders	5,111	Distinct Order IDs
Unique customers	804	Distinct Customer IDs
Product categories	3	Furniture, Office Supplies, Technology
Product sub-categories	17	e.g. Chairs, Phones, Binders
Geographic coverage	4 US regions	West, East, Central, South

4.3 Column Inventory

The 21 columns in the raw dataset cover order identity, customer demographics, geography, product taxonomy, and financial metrics. After column name normalisation (lower-case, spaces replaced with underscores), the key fields used in this project are:

Normalised Column	Type	Description
order_id	String	Unique order identifier
order_date, ship_date	Date	Transaction and fulfilment dates
customer_id, customer_name	String	Customer identity
segment	Categorical	Consumer, Corporate, Home Office
country/region, state/province, city	String	Order delivery geography
region	Categorical	West, East, Central, South
category	Categorical	Furniture, Office Supplies, Technology
sub-category	Categorical	17 product sub-categories
product_name	String	Individual product description
sales	Float	Revenue for the line item (USD)
quantity, discount, profit	Float	Units sold, discount rate (0–1), profit (USD)

5. Data Preprocessing

5.1 Data Loading

The dataset is loaded from `data/raw/sample_-_superstore.xls` using `pandas.read_excel()`. The `data_loader.py` module uses `pathlib` for platform-independent path resolution and searches for the file automatically if it is not found at the exact path. This ensures the pipeline is reproducible across different operating systems and user environments without hardcoded paths.

5.2 Data Quality Assessment

A systematic quality report is generated covering: shape and column inventory, data type classification, missing value counts per column, duplicate row detection, and basic descriptive statistics. The Superstore dataset was found to be high quality with no missing values in any column and no duplicate rows. Date columns (`order_date`, `ship_date`) required explicit dtype coercion from the Excel object type to pandas `datetime64`.

5.3 Column Normalisation

Raw column names contained mixed casing, spaces, and slash characters (e.g. `Country/Region`, `Sub-Category`). All column names were normalised to `lower_snake_case` by converting to lower-case, replacing spaces and slashes with underscores, and removing trailing underscores. This normalisation step is performed before any downstream processing to prevent column-name-not-found errors that would otherwise require conditional logic throughout the codebase.

5.4 Feature Engineering

Eleven derived features were added to a clean copy of the processed dataset. The raw data file was never modified. The processed dataset is stored as a Parquet file at `data/processed/orders_processed.parquet` for efficient downstream loading.

Feature	Formula / Source	Purpose
<code>order_year</code>	<code>order_date.dt.year</code>	Year-level trend analysis
<code>order_month</code>	<code>order_date.dt.month</code>	Monthly seasonality analysis
<code>order_quarter</code>	<code>order_date.dt.quarter</code>	Quarterly aggregations
<code>order_month_name</code>	<code>order_date.dt.month_name()</code>	Human-readable month labels
<code>order_day_of_week</code>	<code>order_date.dt.day_name()</code>	Day-of-week pattern analysis
<code>profit_margin</code>	<code>profit / sales</code>	Line-level profitability ratio
<code>sales_per_quantity</code>	<code>sales / quantity</code>	Effective unit price
<code>is_profitable</code>	<code>profit > 0</code>	Binary flag for filtering
<code>year_month</code>	<code>order_date.dt.to_period('M')</code>	Time-series grouping key
<code>order_size_category</code>	Quartile bins on sales	Order size segmentation
<code>days_to_ship</code>	<code>ship_date - order_date</code>	Fulfilment speed metric

5.5 Data Integrity Decisions

Several records have negative profit values. These are retained as they represent genuinely loss-making transactions (due to high discounts or returns implied by the data structure) and are analytically significant. Removing them would produce misleadingly optimistic

profitability metrics. Profit margin values outside the range [-100%, +100%] can theoretically occur due to extreme discounting; these outliers are retained in the core dataset but noted when they affect aggregate statistics.

6. Exploratory Data Analysis

6.1 Sales Performance Overview

Total revenue across all four years is \$2,326,534.35 from 5,111 unique orders, with an average order value of \$455.20. The portfolio profit is \$292,296.81, representing a portfolio-wide profit margin of 12.56%.

Year	Sales (\$)	Profit (\$)	Profit Margin (%)
2023	494,188	51,717	10.5%
2024	473,085	62,030	13.1%
2025	614,025	82,719	13.5%
2026	745,236	95,831	12.9%
Total	2,326,534	292,297	12.56%

The year-over-year analysis reveals consistent revenue growth from 2023 to 2026, with a particularly strong acceleration in 2025 (+29.8% YoY) and 2026 (+21.4% YoY). Profit margin improved from 10.5% in 2023 to 13.5% in 2025 before a modest decline to 12.9% in 2026, suggesting slight margin pressure as the business scales.

6.2 Category Performance

Category	Sales (\$)	Profit (\$)	Margin (%)
Technology	839,860	145,996	17.4%
Office Supplies	731,868	126,015	17.2%
Furniture	754,806	20,285	2.6%
Portfolio	2,326,534	292,296	12.56%

Technology is the most profitable category both in absolute terms (\$145,996) and margin (17.4%). Office Supplies generates comparable margin (17.2%) from a lower revenue base. Furniture is a major strategic concern: it contributes 32.4% of total revenue but only 6.9% of total profit, with a margin of just 2.6%. This underperformance is driven primarily by the Tables and Bookcases sub-categories.

6.3 Sub-Category Profitability

Sub-category analysis reveals significant profit concentration and loss pockets:

Sub-Category	Sales (\$)	Profit (\$)	Margin (%)
Copiers	150,820	56,085	37.2%
Phones	330,827	44,993	13.6%
Accessories	167,052	41,929	25.1%
Chairs	328,450	26,587	8.1%
Supplies (loss)	46,673	-1,171	-2.5%
Bookcases (loss)	114,880	-3,632	-3.1%
Tables (loss)	207,671	-17,753	-8.5%

Copiers have exceptional margins (37.2%) with moderate volume. Tables destroy the most value in absolute terms (-\$17,753), making them the highest-priority sub-category for

repricing or discount restructuring. The coexistence of high-revenue sub-categories with negative profits (Tables: \$207K revenue but –\$17.7K profit) confirms that revenue growth alone is not a sufficient performance indicator.

6.4 Regional Performance

Region	Sales (\$)	Profit (\$)	Margin (%)
West	739,838	110,979	15.0%
East	691,804	94,832	13.7%
Central	503,246	39,732	7.9%
South	391,646	46,753	11.9%
Total	2,326,534	292,296	12.56%

The West region leads on both revenue (\$739,838) and margin (15.0%). The Central region is a notable underperformer with a margin of only 7.9% despite generating \$503,246 in revenue — over 4 percentage points below the East and 7 points below the West. This suggests pricing, discount, or product mix issues specific to the Central region that warrant targeted investigation.

6.5 Discount Analysis

Discount depth and profitability exhibit a strong negative relationship. The following table shows average profit margin by discount band:

Discount Band	Average Profit Margin (%)	Interpretation
0% (no discount)	34.1%	Highly profitable
1–10%	12.4%	Moderate, still profitable
11–30%	4.7%	Thin margin
31–40%	–12.2%	Loss-making threshold
61–80%	–76.5%	Severely loss-making
Over 80%	–182.4%	Catastrophic margin destruction

The break-even discount threshold appears to lie between 20–30%. Any discount above 30% consistently results in negative margins. The presence of discounts exceeding 80% — which generate margins of –182% — suggests either data entry errors or a fundamentally uncontrolled promotional pricing policy. This finding has direct strategic implications for discount governance.

7. SQL Analysis

7.1 Database Architecture

A SQLite database (superstore.db) is constructed from the processed dataset using Python's sqlite3 module, abstracted through the database.py module. SQLite was chosen because it requires no server installation, is included in Python's standard library, and is fully sufficient for the analytical query volumes involved. The database contains a single denormalised table (orders) mirroring the processed dataset structure, which is appropriate for ad-hoc analytics workloads where join performance is not a concern.

7.2 Business Queries

Twelve business-oriented SQL queries are stored in `sql/business_analysis.sql` with inline comments documenting the business question each query addresses. The queries are designed to be readable and instructive rather than just functional. All queries were executed and validated against the Python-computed equivalents to confirm result consistency.

Query ID	Business Question	Key Insight
Q1	Total sales and profit by category	Furniture lowest margin (2.6%)
Q2	Monthly sales trend	Q4 consistently peaks
Q3	Regional performance comparison	Central region underperforms
Q4	Top 10 customers by revenue	Sean Miller top by sales, unprofitable
Q5	Top 10 products by profit	Technology products dominate
Q6	Loss-making sub-categories	Tables, Bookcases, Supplies
Q7	Average discount by category	Furniture highest avg discount
Q8	Sales and profit by customer segment	Home Office highest margin
Q9	Products with high sales but low profit	Tables, Binders identified
Q10	Year-over-year growth	Consistent revenue growth
Q11	Quarterly performance	Q4 strongest across all years
Q12	Discount impact on profitability	Discount > 30% → losses

8. Customer Analytics

8.1 RFM Methodology

RFM (Recency, Frequency, Monetary) segmentation is a well-established customer analytics framework that classifies customers based on three dimensions of purchasing behaviour:

- **Recency (R):** How recently did the customer last place an order? Customers with more recent purchases are considered more engaged. The reference date is set to the day after the most recent order date in the dataset (2026-12-31 + 1 day) to ensure all customers have a positive recency value.
- **Frequency (F):** How many distinct orders has the customer placed? Higher frequency indicates loyalty and habitual purchasing.
- **Monetary (M):** What is the customer's total revenue contribution? This measures absolute economic value.

Each dimension is scored from 1 (lowest) to 4 (highest) using quartile-based binning, yielding a composite RFM score. Customers are then assigned to one of five business-meaningful segments based on score patterns:

Segment	Customer Count	Segment Definition
Champions	93	High R, F, and M — recent, frequent, high-value
Loyal Customers	203	High F and M — consistent buyers
Potential Loyalists	167	Recent but lower frequency — growth opportunity
At Risk	201	Previously engaged but declining recency
Lost / Inactive	140	Low R, F, M — disengaged customers
Total	804	All unique customers

8.2 RFM Limitations

Several limitations apply to this RFM implementation: (1) The dataset does not contain explicit customer return or churn events, so 'lost' status is inferred from low recency scores only. (2) Quartile-based scoring is sensitive to distribution skewness; the Monetary dimension in particular has a long right tail. (3) The reference date for recency calculation is derived from the dataset itself rather than a true 'today' date, which means the model cannot be applied to real-time scoring without modification. (4) RFM segments are relative within this dataset and should not be compared to benchmark thresholds from other industries or datasets.

8.3 Customer Segment (Business Segment) Analysis

The dataset's native Segment field (Consumer, Corporate, Home Office) provides an additional customer classification layer distinct from RFM. Home Office customers generate the highest profit margin (14.02%) despite having the smallest customer count (149), making them disproportionately profitable per customer.

Segment	Customers	Revenue (\$)	Margin (%)
Consumer	416	1,161,404	11.65%
Corporate	239	706,148	13.17%
Home Office	149	458,982	14.02%
Total	804	2,326,534	12.56%

A noteworthy finding from the top-customer analysis is that the highest-revenue customer (Sean Miller, Home Office segment, \$25,043 revenue) generated a negative profit of -\$1,981, making him the single most unprofitable high-value customer in the dataset. This illustrates the danger of optimising for revenue metrics without profitability visibility.

9. Product Profitability Analysis

Product profitability analysis classifies products along two independent axes — revenue and profit — to identify four strategic quadrants: Stars (high revenue, high profit), Cash Drains (high revenue, low/negative profit), Niche Winners (low revenue, high profit), and Underperformers (low revenue, low profit).

The most strategically important finding is the Cash Drain quadrant: Tables generate \$207,671 in revenue (the 4th highest sub-category) while producing a loss of -\$17,753. This represents an active financial leak at scale. Any growth in Tables sales volume without repricing would increase the absolute loss proportionally.

Copiers represent the ideal Star product: moderate revenue (\$150,820) with exceptional margin (37.2%). Strategic investment in growing Copier sales volume could yield significant profit growth without requiring operational changes.

Chairs illustrate a volume-dependent, margin-compressed product: \$328,450 revenue with only 8.1% margin (\$26,587 profit). These products are important revenue contributors but require volume to justify the operational complexity they introduce.

10. Predictive Analytics

10.1 Forecasting Objective and Data Preparation

Monthly aggregated sales are the forecasting target. The 48 monthly observations spanning January 2023 to December 2026 are transformed into a supervised learning problem by constructing lag features, rolling statistics, and time index features. The feature set includes: month number (1–12), year, quarter, lag-1 through lag-3 sales, rolling 3-month and 6-month mean, and rolling 3-month standard deviation.

10.2 Train/Test Split

A temporal split is used: the training set covers January 2023 to December 2025 (36 months), and the test set covers January 2026 to December 2026 (12 months). This is the only methodologically valid split for time-series data. Random splitting would cause data leakage by allowing the model to train on future observations, which would produce misleadingly optimistic evaluation metrics that would not generalise to real deployment.

10.3 Models and Baseline

Three models were evaluated: a Naive Baseline (predicting next month's sales as equal to the previous month's sales), Linear Regression (sklearn LinearRegression applied to the engineered feature set), and Random Forest Regressor (sklearn RandomForestRegressor with 100 estimators, random_state=42 for reproducibility). Results are evaluated on the held-out 2026 test set:

Model	MAE (\$)	RMSE (\$)	R ² Score	Interpretation
Naive Baseline	—	43,964	-1.84	Worse than mean predictor
Linear Regression	19,926	23,846	0.166	Best performer; limited fit
Random Forest	—	37,636	-1.08	Overfits training set
Target range	~\$30K–\$80K/mo			Context for error magnitudes

10.4 Interpretation of Results

Linear Regression is the best-performing model, achieving a MAE of \$19,926 and RMSE of \$23,846 on the test set. However, an R² of 0.166 indicates the model explains only 16.6% of the variance in monthly sales — a modest fit. The Random Forest and Naive Baseline both produce negative R² scores, meaning they perform worse than simply predicting the mean sales value for every month. These results are honest and are not artificially improved.

The primary cause of the limited forecasting performance is dataset size: 48 monthly observations, of which only 36 are available for training, is insufficient to reliably learn complex temporal patterns. Time-series forecasting typically requires several years of data at the target frequency, or higher-frequency data, to achieve strong predictive performance. Dedicated time-series methods (ARIMA, Prophet, SARIMA) would be more appropriate for this use case with more data, but were not implemented here to avoid overengineering a solution with insufficient data to justify the additional complexity.

10.5 Forecasting Limitations

The following limitations apply: (1) 48 months is a small dataset for machine learning forecasting; confidence intervals cannot be reliably estimated. (2) The model does not

account for external factors (economic cycles, promotional calendars, competitive actions). (3) Monthly aggregation smooths within-month patterns. (4) The Random Forest model exhibits severe overfitting on the training data, consistent with the expectation that tree-based models require substantially more data than is available here. (5) These models should be treated as exploratory baselines, not production forecasting tools.

11. AI Integration

11.1 Design Approach

The AI integration component is implemented in `src/ai_insights.py` and follows a strict data-first design: computed analytical metrics are assembled into a structured context dictionary before any AI component is invoked. The AI receives verified numbers — not raw data — and generates natural-language interpretation of those numbers. This architecture prevents the model from inventing findings and ensures that every AI-generated statement is grounded in computed evidence.

11.2 Operating Modes

The module supports two operating modes:

- **Template/Demo Mode (default):** A structured insight template is populated with the computed metrics to produce deterministic, reproducible business interpretation. This mode requires no external API, no credentials, and no internet connection. It is the default mode used in this project to ensure the submission is fully self-contained and reproducible.
- **IBM Granite Mode (optional):** When the environment variable `USE_GRANITE=1` is set, the module attempts to use an IBM Granite language model via the `ibm-generative-ai` SDK or a local Ollama instance. This mode is explicitly optional and is not required for the project to function. The mode selection is clearly documented in the README and dashboard UI.

11.3 AI Output Structure

The AI output is structured into three sections: Key Findings (a synthesis of the most significant computed metrics), Business Implications (actionable interpretations for management), and Areas for Further Investigation (gaps or anomalies in the data that warrant deeper analysis). All AI-generated output in the dashboard is visually labelled with an 'AI-Generated Interpretation' disclaimer to clearly distinguish it from computed factual evidence. This is a responsible AI design choice: the boundary between verified data and AI interpretation must be explicit and visible.

12. Dashboard Description

The interactive dashboard is built with Streamlit and uses Plotly Express for visualisations. It uses a dark enterprise theme (dark background, orange accent) suitable for business presentations. The dashboard is structured into five pages accessible via the sidebar.

Page	Title	Content
1	Executive Overview	6 KPI cards, sales trend, profit trend, category and regional charts
2	Customer Analytics	RFM segment distribution, segment comparison, top customers, revenue distribution
3	Product & Profitability	Category/sub-category performance, profitability scatter, loss-making products, discount vs profit
4	Sales Prediction	Actual vs predicted chart, model metrics table, methodology explanation, limitations
5	AI Business Insights	Computed metrics panel, AI-generated interpretation, mode indicator, disclaimer
Sidebar	Global Filters	Year, region, category, segment filters with Reset button

All visualisations are generated from actual data at runtime — no charts use hardcoded values. The dashboard caches data loading and preprocessing using `@st.cache_data` to avoid redundant processing on rerender.

13. Results and Key Findings

The following key findings emerge from the complete analysis:

- F1 — Revenue growth is consistent but margin is at risk. Total revenue grew from \$494K (2023) to \$745K (2026), a 51% four-year increase. However, the profit margin declined slightly from 13.5% (2025) to 12.9% (2026), suggesting marginal cost pressure as volume scales.
- F2 — Furniture is a structural profitability problem. At 2.6% margin on \$754K revenue, Furniture contributes 32.4% of revenue but only 6.9% of profit. The Tables sub-category alone destroys \$17,753 in value annually. This is not a market performance issue but a pricing and discount governance issue.
- F3 — Discount policy is the single largest lever for immediate profit improvement. The data shows a clear, monotonic relationship between discount depth and margin destruction. No discount above 30% is profitable on average. Eliminating or capping heavy discounts would have an immediate and substantial impact on portfolio profit.
- F4 — The Central region requires investigation. At 7.9% margin versus a portfolio average of 12.56%, the Central region is the largest margin drag. The root cause is not immediately visible from the dataset alone but may reflect product mix, higher discount rates, or customer segment differences.
- F5 — Customer profitability is not correlated with customer revenue rank. The highest-revenue customer is unprofitable. Without profitability-weighted customer analytics, sales teams may be over-investing in account management for customers that destroy value.
- F6 — Copiers are the highest-priority growth opportunity. With 37.2% margins, any increase in Copier sales volume translates directly to portfolio profit improvement without requiring structural changes.
- F7 — Sales forecasting with 48 monthly observations is limited. The Linear Regression model achieved $R^2=0.166$ on the test set, which is better than both

baselines but still low. This is an honest reflection of data constraints, not a failure of methodology.

14. Business Implications

- I1 — Discount governance policy: Implement a discount approval threshold at 20–25% for all product categories, with explicit executive approval required for any discount exceeding 30%. The current data shows that uncontrolled discounting above this level consistently produces losses.
- I2 — Tables repricing or discontinuation: Tables generate \$207K in revenue and \$17.7K in losses. The business should either reprice Tables to recover margin, negotiate better cost structures with suppliers, or consider discontinuing the sub-category and redirecting customers to profitable alternatives.
- I3 — Technology and Office Supplies investment: Both categories produce 17%+ margins. Shifting marketing investment and sales incentives toward these categories and away from Furniture would improve portfolio-level profitability.
- I4 — At-Risk customer re-engagement: 201 customers are classified as At Risk under the RFM model. A targeted re-engagement campaign (e.g. personalised offers, account review calls) for this segment could recover revenue that would otherwise be permanently lost.
- I5 — Central region operational review: The 4–7 percentage point margin gap between the Central region and all others warrants a dedicated operational review covering pricing, customer mix, and product assortment.

15. Limitations

- Dataset limitations: The Superstore dataset is a demonstration dataset, not actual operational data. While it is structured realistically, it does not capture product costs (only profit), return rates, inventory levels, or competitive context. The discount rates appear to contain some extreme values (>80%) that may not reflect real-world pricing scenarios.
- Forecasting limitations: 48 monthly observations is insufficient for robust time-series modelling. The models produced are exploratory baselines only. Negative R^2 scores on Random Forest and the Naive Baseline reflect genuine data constraints, not modelling errors.
- RFM limitations: Quartile-based scoring is distribution-dependent and cannot be compared across datasets. The 'Lost' segment classification is based entirely on low recency, which does not distinguish between genuinely churned customers and low-frequency buyers who purchase infrequently but are still active.
- AI limitations: The AI insight module in Template/Demo Mode produces deterministic, template-driven output. While grounded in computed metrics, it does not have the reasoning capability of a live LLM and cannot provide novel inferences beyond what the template structure allows. The optional IBM Granite mode provides genuine generative capability but requires external configuration.

16. Responsible AI Considerations

This project was designed with the following responsible AI principles:

- **Transparency:** All AI-generated output is explicitly labelled with a disclaimer. The distinction between computed facts and AI interpretation is maintained throughout the dashboard and this report.
- **Groundedness:** The AI module is architected to receive verified analytical results as context, preventing it from generating findings that are not supported by actual data. The model is not permitted to access raw data directly.
- **Honesty about limitations:** The forecasting models report genuine evaluation metrics including negative R^2 scores. No results are artificially improved or selectively reported.
- **No credential exposure:** The project does not hardcode API keys, passwords, or machine-specific paths. The AI component operates in a credential-free demo mode by default.
- **Reproducibility:** Fixed random seeds (`random_state=42`) are used throughout the machine learning components to ensure all results can be reproduced exactly from a clean environment.
- **Data integrity:** The raw dataset is never modified. All transformations are applied to copies and the lineage from raw data to derived features is fully documented.

17. Conclusion

This project demonstrates a complete, end-to-end data analytics and AI pipeline applied to a realistic retail dataset. Starting from raw Excel data, the project progresses through systematic data quality assessment, feature engineering, multidimensional exploratory analysis, SQL-based business intelligence, customer segmentation, product profitability analysis, time-series forecasting, AI-assisted interpretation, and interactive dashboard delivery.

The key business finding is that the Superstore's profitability challenges are not primarily a revenue problem — total revenue grew consistently at 51% over four years — but a margin management problem rooted in uncontrolled discounting and product mix imbalances. Addressing Tables repricing, enforcing a 20–25% discount ceiling, and reallocating investment toward Technology and Office Supplies represent the three most actionable and highest-impact recommendations the data supports.

The project architecture is modular, documented, and reproducible. Every component is traceable to actual data, every model metric is computed from genuine predictions, and every AI-generated statement is explicitly labelled as interpretation rather than evidence. This approach reflects the standards expected of professional-grade data analytics work in an AI-integrated environment.

18. References

[1] Tableau Software. Sample Superstore Dataset. Available via Tableau Public and packaged workbook. <https://public.tableau.com/>

- [2] McKinney, W. (2010). Data Structures for Statistical Computing in Python. Proceedings of the 9th Python in Science Conference. (pandas library)
- [3] Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.
- [4] Streamlit Inc. (2023). Streamlit — The fastest way to build data apps. <https://streamlit.io/>
- [5] Plotly Technologies Inc. (2023). Collaborative data science — Plotly. <https://plotly.com/>
- [6] Hughes, A. M. (1994). Strategic Database Marketing. Probus Publishing Company. (Original RFM framework reference)
- [7] Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and Practice (3rd ed.). OTexts. <https://otexts.com/fpp3/>
- [8] IBM SkillsBuild. Data Analytics with AI — Academic Internship Program. <https://skillsbuild.org/>
- [9] SQLite Consortium. SQLite — Small. Fast. Reliable. Choose any three. <https://www.sqlite.org/>

Note on AI use: This report was written based on verified analytical results produced by executing the project pipeline. The AI component described in Section 11 uses template-driven interpretation of computed metrics. No AI tool was used to fabricate dataset values, model metrics, or analytical conclusions. All numerical results cited in this report were produced by running the Python pipeline against the actual dataset.